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FINAL YEAR PROJECT

Adaptive Explainable Ensemble with NLP Fusion for Pre-Launch Steam Game Reception Prediction under Varying Feature Availability

PPRS - Chapter 3: Methodology

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Contents

List Of Table

List Of Figures

3.1 Chapter Overview

Methodology shapes how a project is conducted, interpreted and ultimately assessed. A poorly chosen methodological framework can undermine work that is otherwise technically sound, whereas a coherent and well-reasoned approach strengthens every subsequent decision. This chapter sets out the methodological choices made for the Pre-Launch Player Reception System (PPRS), a machine learning system designed to predict whether a Steam game will cross the 75% positive review threshold before release, using only pre-launch developer data.

The chapter covers three areas. Section 3.2 examines the research methodology, using the Research Onion model (Saunders, Lewis and Thornhill, 2023) as a framework for justifying the philosophical and procedural decisions specific to this project. The development methodology governing the PPRS build process is addressed in Section 3.3. Section 3.4 then turns to project management, encompassing scope, scheduling, resource planning and risk identification. A closing summary is provided in Section 3.5.

3.2 Research Methodology

The research methodology for this project is structured around the Research Onion model presented by Saunders, Lewis and Thornhill (2023). The model progresses from broad philosophical assumptions at its outer layers towards specific decisions about data collection and analysis. Table 1 sets out the position taken at each layer, together with the reasoning specific to PPRS.

Research Onion Layer	Selection and Justification for PPRS
Research Philosophy	Positivism is the chosen research philosophy for this project. PPRS operates on factual, historically recorded data from the Steam platform. The prediction target is a binary, measurable outcome: a game either reached the 75% positive review threshold or it did not reach it. This outcome is objective and verifiable, admitting no subjective interpretation. Positivism holds that valid knowledge is derived through observation and quantitative measurement, which corresponds directly with the analytical basis of this study (Saunders, Lewis and Thornhill, 2023).
Research Approach	A deductive approach is taken. The project begins from a defined hypothesis: that structured pre-launch metadata combined with semantic description embeddings can predict Steam game reception. This hypothesis is then tested against an

Research Onion Layer	Selection and Justification for PPRS
	existing dataset through controlled experimentation. The reasoning proceeds from established theory toward empirical evidence. This contrasts with inductive inquiry, where raw observation would precede theory and ideas would emerge from data rather than be tested against it (Saunders, Lewis and Thornhill, 2023).
Research Strategy	Two strategies are combined. The experimental element involves training and evaluating classifiers across five graduated feature tiers, with controlled ablation studies isolating the contribution of individual components, including the SBERT embeddings and the confidence-based routing mechanism. The secondary data analysis element indicates that no primary data collection is undertaken. All analytical work draws on the Ermilov (2025) Steam Games Dataset 2025, a pre-existing Kaggle resource comprising roughly 20,383 filtered records.
Methodological Choice	Mono-method quantitative is the methodological choice for PPRS. The entire project pipeline operates through numerical analysis, from feature preparation through to classifier evaluation. Performance is assessed using F1-score, precision, recall and AUC-ROC. SHAP and LIME produce numerical attribution scores rather than qualitative descriptions. No interviews, surveys or interpretive methods are incorporated, as the prediction target is an objective, recorded outcome that requires no qualitative validation.
Research Time Horizon	A cross-sectional time horizon is adopted. The dataset is a snapshot of Steam game metadata and review outcomes fixed at the point of its compilation. No attempt is made to track how individual titles or broader market conditions evolve across time. All model training, evaluation and explainability analysis are conducted on this single, time-bounded dataset (Saunders, Lewis and Thornhill, 2023).
Research Data Collection and Analysis	Data collection draws entirely on the Ermilov (2025) Steam Games Dataset 2025 sourced from Kaggle. After filtering, the dataset contains roughly 20,383 records, each with 33 pre-launch features. The class distribution is approximately 71.8% positive to 28.2% negative. No primary data is gathered at any stage. Analysis employs Random Forest, CatBoost and XGBoost for binary classification, Sentence-BERT (all-MiniLM-L6-v2) for semantic description encoding with

Research Onion Layer	Selection and Justification for PPRS
	PCA reduction to fifty components, and standard classification metrics alongside SHAP and LIME explainability outputs for evaluation.

Table 1: Research methodology positions based on the Research Onion model (Saunders, Lewis and Thornhill, 2023).

One aspect of this methodology that may invite scrutiny is the absence of surveys or interviews. Some research projects, particularly those involving perceptual output or user experience, genuinely require human participant input to validate results. PPRS falls into a different category. The prediction target is a binary, historically recorded outcome drawn from Steam review data: a game either crossed the reception threshold or it did not do so. This outcome is entirely verifiable from the data, without reliance on human judgment. Introducing surveys would impose a layer of subjectivity upon a framework in which such a dimension is neither warranted nor appropriate.

3.3 Development Methodology

The development methodology governs the process through which PPRS is built, tested and refined. For a project centered on an existing dataset and structured around machine learning experimentation, the chosen model must support iterative cycles, accommodate revision of earlier stages when new findings demand it, and remain practical for a single researcher working independently.

Several candidates were assessed. Waterfall was rejected at the outset. Its phase-locked, sequential structure assumes that requirements are fully determined before work begins and that each stage concludes before the next is entered. Machine learning development does not conform to this pattern. When modelling identifies a weakness in a feature subset, returning to the data preparation stage is a routine necessity rather than an exception. Agile Scrum was also excluded, as it is organized around team sprints, shared backlog reviews and group demonstrations, none of which apply naturally to solo academic research. The Prototyping model supports iteration but is best suited to systems where user reactions to successive interface versions drive refinement. In PPRS, evaluation metrics rather than user responses govern each development cycle.

CRISP-DM (Cross-Industry Standard Process for Data Mining), first formalized by Chapman et al. (2000) and confirmed as a widely applied framework in current data science practice (Martinez-Plumed et al., 2021), is selected as the development methodology for this project. The model defines six phases: Business

Understanding, Data Understanding, Data Preparation, Modelling, Evaluation and Deployment. Each phase corresponds to a recognizable stage in the PPRS pipeline.

Business Understanding encompasses the formulation of the pre-launch reception prediction problem and the design of the five-tier feature availability framework. Data Understanding covers exploratory analysis of the Ermilov (2025) Steam Games Dataset 2025, examining class balance, feature completeness and data quality across all tiers. Data Preparation includes feature engineering, handling of missing values, Sentence-BERT encoding of game descriptions and PCA reduction to fifty components. The Modelling phase involves training and tuning Random Forest, CatBoost and XGBoost classifiers across Models A through E, together with development of the confidence-based routing mechanism. Evaluation encompasses performance measurement using F1-score, precision, recall and AUC-ROC, alongside controlled ablation studies. Deployment corresponds to the Streamlit application, which presents SHAP and LIME explainability outputs alongside each prediction.

CRISP-DM treats its phases as cyclical rather than strictly sequential, formally permitting a return to earlier stages when evaluation results call for revision. This characteristic renders it particularly well-suited to the multi-tier ensemble structure of PPRS, where findings at the evaluation stage may necessitate revisions to data preparation or modelling decisions taken in earlier phases.

3.4 Project Management Methodology

Agile PRINCE2 is adopted as the project management approach for PPRS (AXELOS, 2015). This hybrid framework integrates the structured stage controls found in PRINCE2 with the adaptive flexibility associated with Agile practice. Both properties are relevant to this project. The institutional submission schedule imposes fixed, externally defined deadlines for each deliverable, requiring disciplined planning and clear governance. Simultaneously, machine learning development is inherently exploratory. Modelling and evaluation cycles do not follow a predetermined sequence, and the work regularly necessitates adjustment in response to experimental findings. Agile PRINCE2 accommodates both requirements within one coherent framework, making it well-suited to a project that must balance formal academic governance with the open-ended nature of technical experimentation.

3.4.1 Project Scope

PPRS is bounded to predicting pre-launch Steam game reception using only developer-provided metadata available prior to public release. The system accepts structured inputs across five graduated feature tiers and produces a binary classification indicating whether a given title is likely to achieve 75% or more positive user reviews. All modelling and evaluation draws exclusively on the Ermilov (2025) Steam Games

Dataset 2025; no live data is retrieved from the Steam platform at any stage. The system does not incorporate post-launch attributes, does not extend to distribution platforms other than Steam, and does not undertake any primary data collection. The complete scope definition, including a detailed account of what falls within and outside the project boundaries, is provided in Section 1.12 of Chapter 1.

3.4.2 Schedule

3.4.2.1 Gantt Chart

The full project schedule is presented as a Gantt chart in Appendix A. The chart spans July 2026 through April 2027, mapping all formative submission points, thesis chapter writing phases and application development stages across the project timeline.

3.4.2.2 Deliverables and Dates

Table 2 lists all project deliverables with their submission dates and their classification as formative or summative. Summative entries carry formal assessment weighting and are shown in bold. All deadlines are as defined by the Informatics Institute of Technology, Sri Lanka affiliated with University of Westminster, United Kingdom.

Deliverable	Deadline	Type
Introduction Chapter	2 July 2026	Formative
Methodology Chapter	16 July 2026	Formative
Literature Review Chapter	17 September 2026	Formative
Software Requirements Specification (SRS)	15 October 2026	Formative
PPRS - Document and Video Presentation	**19 November 2026**	**Summative (15%)**
Proof of Concept	17 December 2026	Formative
Design and Prototype	21 January 2027	Formative
IPD - Document and Video Demonstration	**11 February 2027**	**Summative (15%)**
Testing and Evaluation	4 March 2027	Formative
First Version of Final Thesis Draft	18 March 2027	Formative
Final Submission - Thesis, Video Demo and Code	**1 April 2027**	**Summative (70%)**

Table 2: Project deliverables, submission deadlines and assessment classification.

3.4.3 Resource Requirements

The following subsections detail the resources required for the successful completion of this project across four categories: hardware, software, data and skills.

3.4.3.1 Hardware Requirements

- A GPU with CUDA support, such as an NVIDIA GTX 1080 Ti or a more recent equivalent, for model training, SBERT encoding and SHAP value computation. Where a suitable local GPU is unavailable, cloud computing environments including Google Colab and Kaggle Notebooks are appropriate substitutes.
- Intel Core i5 8th generation processor or AMD Ryzen 5 equivalent, or above, to support the development environment and data pipeline execution.
- A minimum of 8 GB RAM to support dataset loading and feature engineering operations; 16 GB is advisable where multiple training processes run concurrently.
- Approximately 5 GB of available disk storage, sufficient for the dataset files, trained model artefacts, cached SBERT embeddings and intermediate pipeline outputs.

3.4.3.2 Software Requirements

- Operating system: Windows 11, Ubuntu 22.04 LTS or macOS Ventura (version 13) or later, each providing a supported environment for Python-based machine learning tooling.
- Python 3.10 or later, as the primary programming language for all data processing, model development and deployment tasks.
- Random Forest, CatBoost and XGBoost, comprising the classifier base across Models A through E, implemented via scikit-learn and their respective dedicated libraries.
- Sentence-Transformers library (all-MiniLM-L6-v2), for generating semantic embeddings from game description text.
- Scikit-learn, for PCA dimensionality reduction, data preprocessing and evaluation metric computation.
- SHAP and LIME libraries, for structured feature attribution scores and word-level explainability outputs respectively.
- Streamlit, for building and deploying the interactive prediction and explainability application.
- Jupyter Notebook or an equivalent IDE such as VS Code or PyCharm, for development, experimentation and documentation.

- GitHub, for version control and project history management throughout the development period.

3.4.3.3 Data Requirements

- Ermilov (2025) Steam Games Dataset 2025 (Kaggle). This pre-existing resource contains approximately 20,383 filtered game records with 33 pre-launch metadata features and binary reception labels. It is the sole data source used for model training, evaluation and explainability analysis throughout this project.

3.4.3.4 Skill Requirements

- Machine learning and data science: proficiency in gradient-boosted classifiers, tabular feature engineering, model evaluation and hyperparameter optimization.
- Natural language processing: working familiarity with transformer-based sentence embedding models, dimensionality reduction methods and the integration of text-derived features into structured classification pipelines.
- Explainable AI: understanding of SHAP TreeExplainer methods and LIME perturbation-based explanations for classification models.
- Python programming: competence across the data science stack, covering data handling, model development and application deployment.
- Academic writing: ability to produce well-referenced, technically accurate documentation conforming to institutional standards.
- Communication: capacity to engage constructively with the project supervisor and incorporate feedback at each stage of the project.

3.4.4 Risks and Mitigation

Table 3 presents the principal risks identified in relation to this project. Severity (S) and Likelihood (L) are each rated on a scale of one to five, where five represents the highest level of concern. A mitigation strategy is provided for each identified risk.

S: Severity (1-5) | L: Likelihood (1-5)

Risk Item	S	L	Mitigation Plan
Class imbalance suppressing recall. The dataset has a 71.8%/28.2% class split. This may push classifiers toward the majority class, reducing	4	4	Apply class weighting within CatBoost and XGBoost during training. Prioritise F1-score and AUC-ROC as evaluation metrics, given

Risk Item	S	L	Mitigation Plan
their capacity to correctly identify negative reception cases.			their robustness under imbalanced conditions relative to raw accuracy.
Feature leakage from post-launch attributes. Some fields that appear pre-launch in the dataset, such as estimated owner counts, may in practice reflect post-release figures, contaminating the training signal.	5	3	Conduct a thorough feature audit prior to modelling. Each attribute will be verified against the PPRS pre-launch definition. Any field found to reflect post-launch state will be removed from all feature tiers.
Low-quality embeddings from short or generic descriptions. Titles with minimal or templated text may yield SBERT encodings that introduce noise into Model E rather than useful predictive signal.	3	4	Set a minimum character threshold for game descriptions during data preparation and remove records that fall below it. Ablation studies comparing Model D and Model E will determine whether the embeddings provide a net benefit.
Routing instability near tier boundaries. The confidence-based mechanism directing inputs to Models A through E may produce inconsistent assignments when a record sits close to a confidence score threshold. When two or more eligible specialists return similar confidence values, a minor variation in the input data may cause the router to select a different model, reducing prediction consistency for records that sit near the boundary of model preference.	4	3	Define and document fixed routing thresholds for each tier prior to implementation. Test the routing mechanism using boundary-case inputs during the evaluation phase.
Insufficient local compute for training and SHAP computation. Running five model tiers with SBERT encoding and SHAP value generation may exceed available hardware capacity, causing timeline delays.	3	4	Use cloud-based environments such as Google Colab or Kaggle Notebooks where local resources are inadequate. Apply PCA reduction to fifty components early in the pipeline to limit memory demands from SBERT encoding.

Risk Item	S	L	Mitigation Plan
Slow SHAP and LIME computation within the Streamlit application. Real-time explainability generation may be too computationally demanding for a responsive user interface.	3	3	Pre-compute SHAP values for representative sample inputs during development. Reduce LIME perturbation sample size. Apply output caching within the Streamlit application where technically feasible.
Scope growth during development. The five-tier architecture may expand as additional feature variants or model alternatives are explored, placing the project schedule at risk.	4	3	Adhere to the scope boundaries defined in Section 1.12. Any proposed additions will be reviewed with the project supervisor before implementation proceeds. The Agile PRINCE2 governance structure provides the change-control mechanism for this.
Dataset quality concerns. The Ermilov (2025) dataset may contain inaccurate metadata, missing values at scale, or may not fully represent the current Steam catalogue.	3	3	Conduct thorough exploratory data analysis and quality checks before modelling begins. All cleaning decisions will be documented. Dataset limitations will be stated explicitly in the final thesis.

Table 3: Identified project risks, severity and frequency ratings, and mitigation strategies.

3.5 Chapter Summary

This chapter has presented the complete methodological framework for PPRS across its research, development and project management dimensions.

With regard to research methodology, positivism was established as the guiding philosophy, grounded in the objective and measurable character of the prediction task. A deductive approach was taken, testing a stated hypothesis against an existing dataset rather than deriving theory from raw observation. The methodological choice of mono-method quantitative reflects the fact that every stage of the pipeline produces and analyses numerical data. The research strategy combines controlled experimentation with secondary data analysis of the Ermilov (2025) Steam Games Dataset 2025. A cross-sectional time horizon applies, since the dataset represents a fixed point in time. The exclusion of surveys and interviews was justified by the verifiable, non-subjective character of the prediction target.

For development, CRISP-DM was selected because its six phases correspond closely to the PPRS build sequence and because it formally permits iteration between stages when evaluation results require it. Agile PRINCE2 (AXELOS, 2015) was adopted for project management, providing a framework that accommodates both the fixed institutional deadline structure and the adaptive demands of machine learning experimentation. The project scope was cross-referenced to Section 1.12. A full schedule and deliverables table were presented, resource requirements were detailed across four categories, and eight risks were identified and paired with actionable mitigation plans.

Chapter 4 proceeds to the Software Requirements Specification, building on the methodological foundations established here.

References

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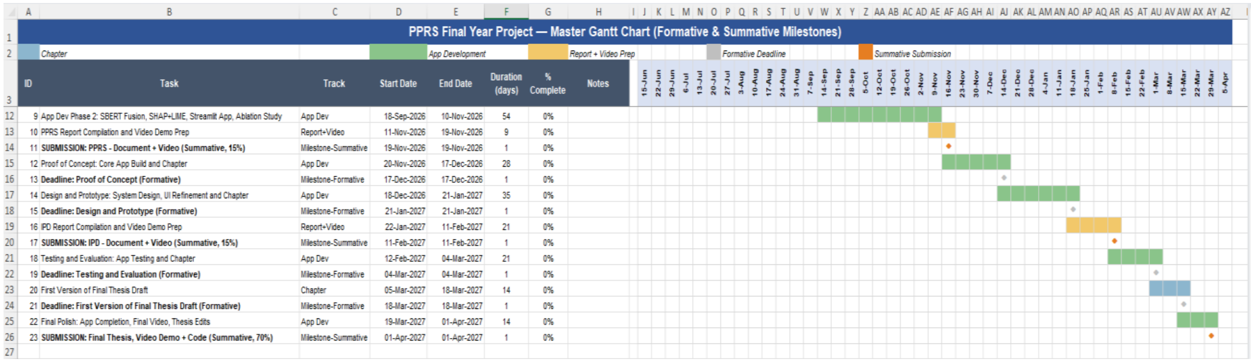
Appendix A



ID	Task	Track	Start Date	End Date	Duration (days)	% Complete	Notes
9	App Dev Phase 2: SBERT Fusion, SHAP+LIME, Streamlit App, Ablation Study	App Dev	18-Sep-2026	10-Nov-2026	54	0%	
10	PPRS Report Compilation and Video Demo Prep	Report+Video	11-Nov-2026	19-Nov-2026	9	0%	
11	**SUBMISSION: PPRS - Document + Video (Summative, 15%)**	Milestone-Summative	19-Nov-2026	19-Nov-2026	1	0%	
12	Proof of Concept: Core App Build and Chapter	App Dev	20-Nov-2026	17-Dec-2026	28	0%	
13	**Deadline: Proof of Concept (Formative)**	Milestone-Formative	17-Dec-2026	17-Dec-2026	1	0%	
14	Design and Prototype: System Design, UI Refinement and Chapter	App Dev	18-Dec-2026	21-Jan-2027	35	0%	
15	**Deadline: Design and Prototype (Formative)**	Milestone-Formative	21-Jan-2027	21-Jan-2027	1	0%	
16	IPD Report Compilation and Video Demo Prep	Report+Video	22-Jan-2027	11-Feb-2027	21	0%	
17	**SUBMISSION: IPD - Document + Video (Summative, 15%)**	Milestone-Summative	11-Feb-2027	11-Feb-2027	1	0%	
18	Testing and Evaluation: App Testing and Chapter	App Dev	12-Feb-2027	04-Mar-2027	21	0%	
19	**Deadline: Testing and Evaluation (Formative)**	Milestone-Formative	04-Mar-2027	04-Mar-2027	1	0%	
20	First Version of Final Thesis Draft	Chapter	05-Mar-2027	18-Mar-2027	14	0%	
21	**Deadline: First Version of Final Thesis Draft (Formative)**	Milestone-Formative	18-Mar-2027	18-Mar-2027	1	0%	
22	Final Polish: App Completion, Final Video, Thesis Edits	App Dev	19-Mar-2027	01-Apr-2027	14	0%	
23	**SUBMISSION: Final Thesis, Video Demo + Code (Summative, 70%)**	Milestone-Summative	01-Apr-2027	01-Apr-2027	1	0%	

Figure 1: Gantt Chart